

o Aim:

- To verify the relation between frequency (n) and resonating volume (V) and hence find frequency of unknown tuning fork.

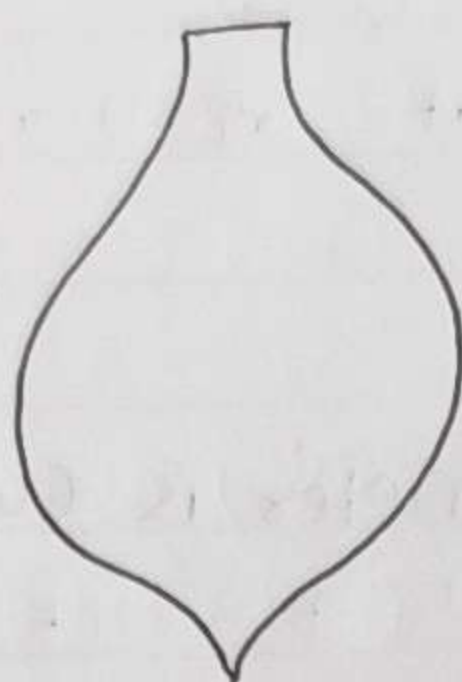
o Apparatus:

- Resonator bottle, tuning forks, rubber pad, measuring cylinder.

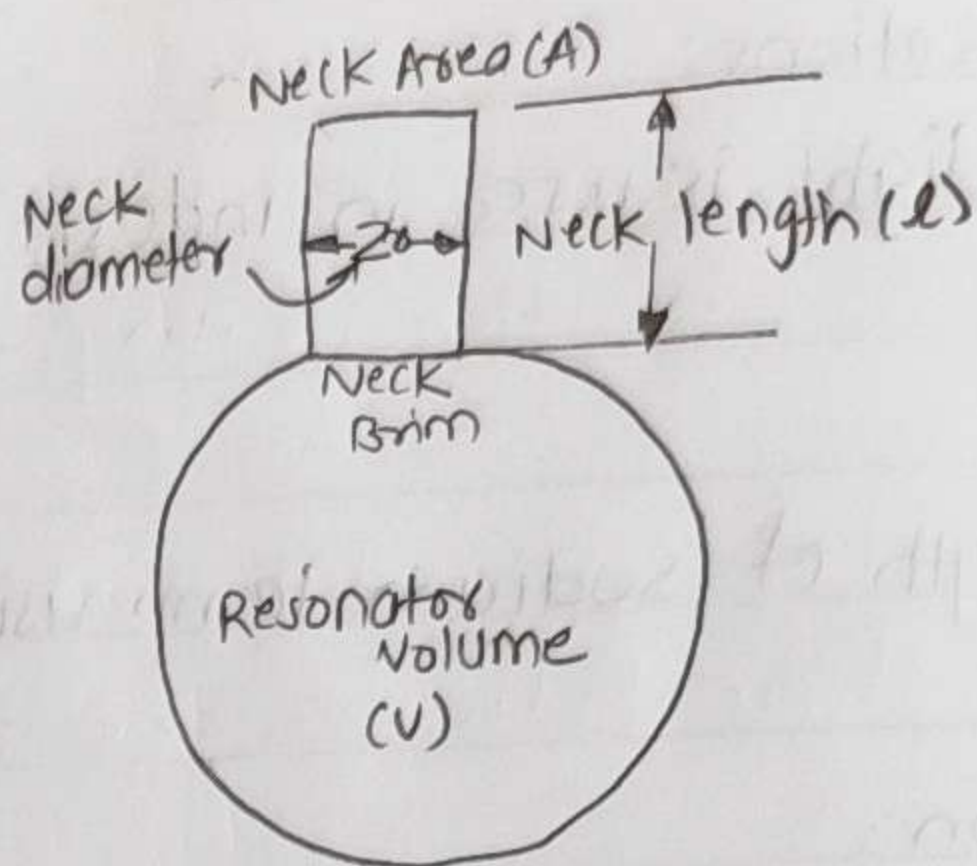
o Theory:

- Hermann von Helmholtz's study of sound is one of the most important contribution to Physics. He studies various combinations of vibrations, beats, their inter-relationships. His study of resonators is the most important amongst these. A coke bottle is a simple volume resonator which makes a unique sound when air is blown into it.
- A resonator is essentially a gas volume having a natural frequency of vibration. It resounds energetically to a tuning fork of same frequency.
- If ' n ' is frequency of tuning and ' V ' is resonating air volume, then for a practical resonator $n^2 V = \text{constant}$, and a graph plotted between V and ' $1/n^2$ ' will be a straight line.
- Helmholtz resonator is a vessel with a narrow neck. If air is blown into it, the air gets compressed and expanded. Instead, one can make the air in the neck portion to move by the sound wave created by placing a tuning fork above the neck.

Diagrams:



Helmholtz Resonator



observations:

Sr. No.	Frequency 'n' (Hz)	Volume 'V' (ml)	$\frac{1}{n^2}$ (s ²)	$n^2 V$ (Hz ² ml)
1	512	100	3.814×10^{-6}	26214400
2	480	129	4.340×10^{-6}	29721600
3	426.6	156	5.494×10^{-6}	28390059.36
4	384	186	6.781×10^{-6}	27426816
5	341.3	235	8.584×10^{-6}	27374137.15
6	320	269	9.765×10^{-6}	27545600
7	288	333	12.056×10^{-6}	27620352
8	256	423	15.258×10^{-6}	27721728
Unknown	n	125	$\frac{1}{n^2}$	$125n^2$

• Calculations:

$$(i) n_1^2 v_1 = (512)^2 (100) = 26214400 \text{ Hz}^2 \text{ m}$$

$$(ii) n_2^2 v_2 = (480)^2 (129) = 29721600 \text{ Hz}^2 \text{ m}$$

$$(iii) n_3^2 v_3 = (426.6)^2 (156) = 28390059.36 \text{ Hz}^2 \text{ m}$$

$$(iv) n_4^2 v_4 = (384)^2 (186) = 27426816 \text{ Hz}^2 \text{ m}$$

$$(v) n_5^2 v_5 = (301.3)^2 (235) = 27374137.15 \text{ Hz}^2 \text{ m}$$

$$(vi) n_6^2 v_6 = (320)^2 (269) = 27545600 \text{ Hz}^2 \text{ m}$$

$$(vii) n_7^2 v_7 = (288)^2 (333) = 27620352 \text{ Hz}^2 \text{ m}$$

$$(viii) n_8^2 v_8 = (256)^2 (423) = 27721728 \text{ Hz}^2 \text{ m}.$$

$$- (n^2 v)_{\text{avg}} = \frac{26214400 + 29721600 + 28390059.36 + 27426816 + 27374137.15 + 27545600 + 27620352 + 27721728}{8}$$

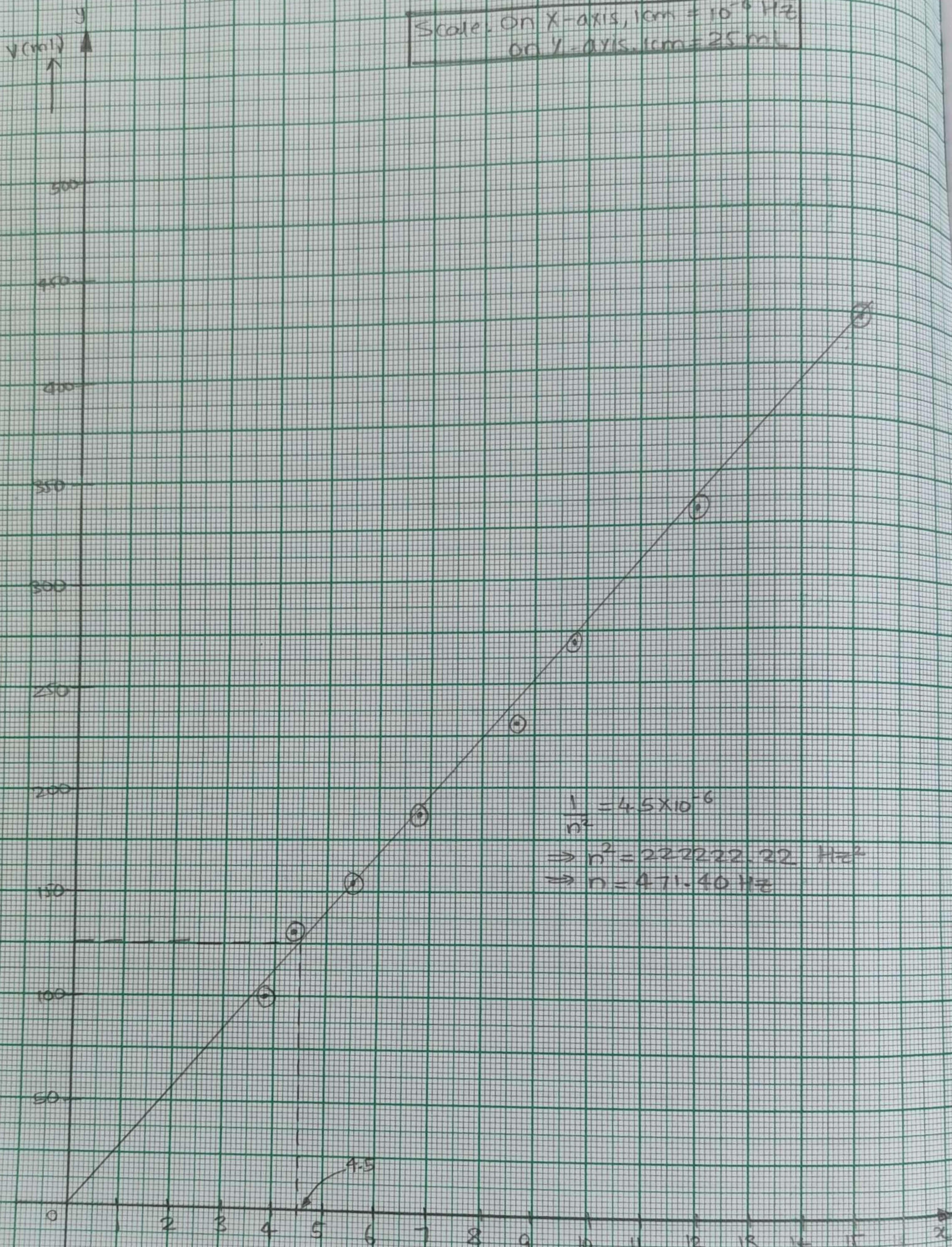
$$\therefore (n^2 v)_{\text{avg}} = \frac{222014692.5}{8} = 27751838.56 \text{ Hz}^2 \text{ m}$$

$$- \text{Now, } n_x = \left[\frac{(n^2 v)_{\text{avg}}}{v} \right]^{1/2} = \left[\frac{27751838.56}{125} \right]^{1/2} = 471.1843 \text{ Hz}$$

$$- \text{Error} = \left| \frac{n_x - n}{n} \right| \times 100 = \left| \frac{480 - 471.1843}{480} \right| \times 100 = 1.836\%$$

- Helmholtz Resonator do not work on the principle of standing waves; instead it can be modeled as a spring-mass system where air in the neck acts like the mass and air pressure acts like the spring.
- o Procedure:
 - Fill the resonator with water upto the neck.
 - Arrange given tuning forks in descending order. Take the tuning fork of highest frequency and strike it against the rubber pad.
 - The vibrating tuning fork is hold near the mouth of the resonator and water is allowed to flow out of the bottle.
 - At resonance, the sound produced will be maximum. The volume of air column is equal to the volume of water dropped out.
 - Repeat the steps for lower frequency values. Calculate the average value of the product n^2V for each frequency n .
 - The unknown tuning fork frequency can be found by:
$$n_x^2 V_x = (n^2 V)_{\text{mean}} \Rightarrow n_x = \left[\frac{(n^2 V)_{\text{mean}}}{V_x} \right]^{1/2}$$
and also find the % error in frequency.
 - Plot the graph of V vs $1/n^2$ and find the tuning fork frequency. Verify that both results are approximately the same.

Scale: On X-axis, 1cm = 10^{-6} Hz
On Y-axis, 1cm = 25 mL



$$\frac{1}{n^2} = 4.5 \times 10^{-6}$$

$$\Rightarrow n^2 = 222222.22 \text{ Hz}^2$$

$$\Rightarrow n = 471.40 \text{ Hz}$$

$\rightarrow \frac{1}{n^2} (\times 10^6 \text{ Hz})$

• Precautions:

- Take care at eye level and avoid parallax error.
- Pour water completely and ensure that stray drops aren't collected.

• Applications:

- Volume resonators are used in sound absorption.
- They are also used in instruments like violin, guitar, etc.

• Result:

- Unknown tuning fork frequency by calculations = 471.18 Hz
- Unknown tuning fork frequency by graph = 471.40 Hz

• Conclusion:

- The relation $n^2V = \text{constant}$ is verified as the plot between V and $1/n^2$ is a straight line.
- The frequency of unknown tuning fork is found out to be 471.18 Hz which has a 1.836% error from the actual value of 480 Hz.